

GAL Seminar 5

ex 1

$$\text{Fie } f: \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad f(x) = (x_1 - x_2 - x_3, x_1 + x_2 - x_3)$$

a) Scrieti mat. lui f in bazele canonice

$$[f] = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \end{pmatrix}$$

" " "

$f(1,0,0)$ $f(0,1,0)$ $f(0,0,1)$

b) Det mat. lui f in raport cu bazele B_1, B_2

unde $B_1 = \{ (1, 1, 1), (2, 2, 1), (3, 1, 2) \}$

$$B_2 = \{ (1, 2), (3, 1) \}$$

c) $[f]_{B_1, B_2'}$ unde $B_2' = \{ (-1, 1), (-1, 3) \}$

ex 2

$$\text{Fie } f: \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad f(x) = (x_1 + x_2 - x_3, x_1 - x_2 - x_3, x_1 + x_2 + x_3)$$

a) Calculati matricea lui f in raport cu baza canonica si $B_1 = \{ (1, 1, 1), (2, 2, 1), (2, 1, 2) \}$

$$f: V \rightarrow W \quad \text{op. liniară}$$

$$B_1 = \{ e_1, e_2, \dots, e_n \} \text{ în } V$$

$$B_2 = \{ g_1, g_2, \dots, g_m \} \text{ în } W$$

$$x = x_1 e_1 + \dots + x_n e_n$$

$$f(x) = x_1 f(e_1) + x_2 f(e_2) + \dots + x_n f(e_n)$$

$$= y_1 g_1 + y_2 g_2 + \dots + y_m g_m = y$$

$$f(e_1) = a_{11} g_1 + \dots + a_{m1} g_m$$

...

$$f(e_j) = a_{1j} g_1 + \dots + a_{mj} g_m \quad \forall j = \overline{1, n}$$

$$\begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

$$[f]_{B_1, B_2}$$

$$[y]_{B_2} = [f]_{B_1, B_2} \cdot [x]_{B_1}$$

$$[x]_{B_1'} = A [x]_{B_1}$$

$$B_1' = \{ e_1', \dots, e_n' \}$$

$$[f]_{B_1', B_2} = [f]_{B_1, B_2} A$$

$$B_2' = \{ g_1', \dots, g_m' \}$$

$$[f]_{B_1', B_2'} = B^{-1}$$

$$B_1 \xrightarrow{A} B_1'$$

$$B_2 \xrightarrow{B} B_2'$$

ex 1

$$\text{Fie } f: \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad f(x) = (x_1 - x_2 - x_3, x_1 + x_2 - x_3)$$

a) Scrieti mat. lini f in bazele canonice

$$[f] = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \end{pmatrix}$$

" " "

$f(1,0,0)$ $f(0,1,0)$ $f(0,0,1)$

b) Det mat. lini f în raport cu bazele B_1, B_2

unde $B_1 = \{ (1, 1, 1), (2, 2, 1), (3, 1, 2) \}$

$$B_2 = \{ (1, 2), (3, 1) \}$$

Baza 1

$$f(1, 1, 1) = (-1, 1)$$

$$f(2, 2, 1) = (-1, 3)$$

$$f(3, 1, 2) = (0, 2)$$

$$\begin{pmatrix} -1 & -1 & 0 \\ 1 & 3 & 2 \end{pmatrix} = [f]_{\text{în } B_1, \text{ canonică}}$$

Baza 2

$$\left(\begin{array}{cc|ccc} 1 & 3 & -1 & -1 & 0 \\ 2 & 1 & 1 & 3 & 2 \end{array} \right)$$

$$\left(\begin{array}{cc|ccc} & & -1 & -1 & 0 \\ & & 4 & 6 & 2 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{cc|ccc} 1 & 3 & -1 & -1 & 0 \\ 0 & 1 & -\frac{1}{2} & -\frac{3}{2} & -\frac{1}{4} \end{array} \right)$$

$$\rightarrow \left(\begin{array}{cc|ccc} 1 & 0 & \frac{1}{2} & \frac{7}{2} & \frac{3}{4} \\ 0 & 1 & -\frac{1}{2} & -\frac{3}{2} & -\frac{1}{4} \end{array} \right)$$

$$[f]_{\beta_1, \beta_2} = \begin{pmatrix} \frac{1}{2} & \frac{7}{2} & \frac{3}{4} \\ -\frac{1}{2} & -\frac{3}{2} & -\frac{1}{4} \end{pmatrix}$$

c) $[f]_{\beta_1, \beta_2'}$ unter $\beta_2' = \{(-1, 1), (-1, 3)\}$

$$\beta_2' = \langle g_1, g_2 \rangle$$

$$\left(\begin{array}{cc|ccc} -1 & -1 & -1 & -1 & 0 \\ 1 & 3 & 1 & 3 & 2 \end{array} \right) \rightarrow$$

$$\rightarrow \left(\begin{array}{cc|ccc} \boxed{1} & 1 & 1 & 1 & 0 \\ 1 & 3 & 1 & 3 & 2 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{cc|ccc} 1 & 1 & 1 & 1 & 0 \\ 0 & 2 & 0 & 2 & 2 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{cc|ccc} 1 & 1 & 1 & 1 & 0 \\ 0 & \boxed{1} & 0 & 1 & 1 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{cc|ccc} 1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 1 & 1 \end{array} \right)$$

ex 2

Fi $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $f(x) = (x_1 + x_2 - x_3, x_1 - x_2 - x_3, x_1 + x_2 + x_3)$

a) Calculati matricea lui f în raport cu baza canonică și $B_1 = \{ (1, 1, 1), (2, 2, 1), (2, 1, 2) \}$

$$[f] = \begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & -1 \\ 1 & 1 & 1 \end{pmatrix}$$

$$f(1, 1, 1) = (1, -1, 3)$$

$$f(2, 2, 1) = (3, -1, 5)$$

$$f(2, 1, 2) = (1, -1, 5)$$

$$\Rightarrow \begin{pmatrix} 1 & 3 & 1 \\ -1 & -1 & -1 \\ 3 & 5 & 5 \end{pmatrix}$$

$\begin{matrix} | & | & | \\ f[g_1] & f[g_2] & f[g_3] \end{matrix}$

$$\left(\begin{array}{ccc|ccc} \boxed{1} & 2 & 2 & 1 & 3 & 1 \\ 1 & 2 & 1 & -1 & -1 & -1 \\ 1 & 1 & 2 & 3 & 5 & 5 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 3 & 1 \\ 0 & 0 & -1 & -2 & -4 & -2 \\ 0 & -1 & 0 & 2 & 2 & 4 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 3 & 1 \\ 0 & \boxed{1} & 0 & -2 & -2 & -4 \\ 0 & 0 & 1 & 2 & 4 & 2 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 5 & 7 & 9 \\ 0 & 1 & 0 & -2 & -2 & -4 \\ 0 & 0 & 1 & 2 & 4 & 2 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 5 \\ 0 & 1 & 0 & -2 & -2 & -4 \\ 0 & 0 & 1 & 2 & 4 & 2 \end{array} \right)$$

$$[4]_{B_1} = \left(\begin{array}{ccc} 1 & -1 & 5 \\ -2 & -2 & -4 \\ 2 & 4 & 2 \end{array} \right)$$